LoRA: Low-Rank Adaptation of Large Language Models

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen (Microsoft

ABSTRACT

An important paradigm of natural language processing consists of large-scale pre-training on general domain data and adaptation to particular tasks. Full fine-tuning becomes prohibitive as models scale to hundreds of billions of parameters. We propose Low-Rank Adaptation (LoRA), which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer

1. Introduction and The Problem of Adaptation Scale

Natural language processing has undergone a revolution driven by scaled pre-trained models such as GPT-3 (175B parameters). Adapting these models via full fine-tuning requires updating all parameters for every downstream task, requiring enormous storage and compute. For GPT-3, storing independent fine-tuned checkpoints of 175B parameters for dozens of tasks is practically unfeasible.

Existing parameter-efficient methods such as prefix tuning or adapter layers introduce non-trivial inference latency (by adding serial computation layers) or reduce the sequence length available for context.

2. Methodology: Low-Rank Matrix Decomposition

We take inspiration from intrinsic dimension studies showing that parameter updates Delta_W during task adaptation reside in a low-dimensional subspace. For a pre-trained weight matrix W_0 in R^{d x k}, we constrain its update by representing it as Delta_W = B * A, where B in R^{d x r} and A in R^{r x k}, with rank r << min(d, k).

During training, W_0 is frozen and receives no gradient updates, while A and B contain trainable parameters. For input x, modified forward pass is h = W_0 * x + (alpha / r) * B * A * x. We initialize A with Gaussian distribution and B with zero, so Delta_W = 0 at the start of training.

Key Advantages: No added inference latency because W = W_0 + Delta_W can be pre-computed and merged directly when deploying. Furthermore, switching between tasks requires swapping only the low-rank matrices.

3. Results on GPT-3 175B, Benchmarks & Limitations

Experiments on RoBERTa, DeBERTa, and GPT-3 175B demonstrate that LoRA performs on par or better than full fine-tuning on GLUE, MultiNLI, and WikiSQL benchmarks, while using 10,000 times fewer trainable parameters and 3 times less GPU memory.

On GPT-3 175B, LoRA reduces trainable parameter count from 175,255.8M to 37.7M (with r = 4), achieving 73.4% accuracy on WikiSQL compared to 73.8% for full fine-tuning.

Limitations: It is not straightforward to batch inputs across different tasks with different rank adaptations in a single forward pass without dynamic kernel dispatching. Additionally, selecting optimal rank r across heterogeneous task families requires empirical tuning.